Program Complexity

"Ideal Complexity"

Serial Programs

1 Instructions

1.1 Definition of Instruction

An instruction is either a canonical comparison, canonical operation, or canonical measurment of framework $\mathcal{F}$

1.2 Definition of Instruction Set

Define $\mathcal{I}$; an ordered set of instructions s_i

$$\mathcal{I} := \{s_1, s_2, ..., s_N\}$$

2 Memory

2.1 Definition of Memory Set

Define Memory $\mathcal{M}$; an ordered set of objects m_i

$$\mathcal{M} := \{m_1, m_2, ..., m_M\}$$

3 Programs

3.1 Definition of a Program

Define Program P

$$X_n = \{x_1, ..., x_n\}$$

$$\mathcal{P} := P[X_n] \rightarrow y_o \in \Omega \quad \forall X_n \in \mathbb{D}_P =$$
$$\{s_1, s_2, ..., s_N \,|y_o, b_2, b_3, ..., b_M\} \rightarrow y_o \in \Omega \quad \forall X_n \in \mathbb{D}_P$$

3.2 Void Programs

Define a void program; a program with inputs x_i and no output

$$X_n = \{x_1, ..., x_n\}$$

$$\mathcal{P} := P[X_n] \quad \forall X_n \in \mathbb{D}_P =$$
$$\{s_1, s_2, ..., s_N \mid b_1, b_2, ..., b_M\} \quad \forall X_n \in \mathbb{D}_P$$

3.3 Boolean Programs

Define a Boolean Program
Given

$$X_n = \{x_1, ..., x_n\}$$

$$\mathcal{P} := P[X_n] \rightarrow y_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_n \in \mathbb{D}_P \quad =$$
$$\{s_1, s_2, ..., s_N \mid b_2, b_3, ..., b_M, y_o\} \rightarrow y_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_n \in \mathbb{D}_P$$

3.4 Numerical Programs

Define a numerical program; a program with inputs x_i, input set C, and real, rational output y_o

$$X_n = \{x_1, ..., x_n\}$$
$$\mathcal{P} := P[X_n] \rightarrow y_o \in \mathbb{Q} \wedge y_o \geqslant 0 \quad \forall X_n \in \mathbb{D}_P =$$
$$\{s_1, s_2, ..., s_N \mid b_1, b_2, ..., b_M, y_o\} \rightarrow y_o \in \mathbb{Q} \wedge y_o \geqslant 0 \quad \forall X_n \in \mathbb{D}_P =$$

3.5 System Programs

Define a system program; a program with inputs x_i, input set C, and real, output set Y_o

$$X_n = \{x_1, ..., x_n\}$$
$$\mathcal{P} := P[X_n] \rightarrow (Y_o \in \Omega = \{y_1, y_2, ..., y_K\} \in \Omega) \quad \forall X_n \in \mathbb{D}_P \quad =$$
$$\{s_1, s_2, ..., s_N \mid b_1, b_2, ..., b_M, Y_o\} \rightarrow (Y_o \in \Omega = \{y_1, y_2, ..., y_K\} \in \Omega) \quad \forall X_n \in \mathbb{D}_P$$

4 Program Complexity

4.1 Definition

Define simple computational complexity of dimension 2

$$\mathbf{O}[n] := \ < O_T[n], O_S[n] >$$

4.2 Total Complexity

Define Total Complexity of solution s^+

$$O[n] := |s^+[X_n]| = |\{\mathcal{L}, \mathcal{M}, y_o\}|$$

$$= |\mathcal{L}| + |\mathcal{M}| + |y_o| = N + M + 1$$

4.3 Time Complexity

Restate definition of Time Complexity $O_T[n]$ of solution s^+

$$s^+ = \{\mathcal{L}, \mathcal{M}, y_o\}$$

$$O_T[n] := |\mathcal{L}| = N$$

4.4 Space Complexity

Restate definition of Time Complexity $O_S[n]$ of solution s^+

$$s^+ = \{\mathcal{L}, \mathcal{M}, y_o\}$$

$$O_S[n] := |\mathcal{M}| + |y_o| = M + 1$$

4.5 Total Complexity as a Function of Time and Space Complexity

$$O[n] := |s^+[X_n]| = |\{\mathcal{L}, \mathcal{M}, y_o\}|$$

$$= |\mathcal{L}| + |\mathcal{M}| + |y_o|$$

$$= O_T[n] + O_S[n]$$

4.6 $O_S[n] > 0^*$

Assuming Program is not void

4.6.1 Proof

Assume $O_S[n] = 0$

$$O_S[n] := |\mathcal{M}| + |y_o|$$

$$O_S[n] = 0 \Rightarrow \mathcal{M} = y_o = \varnothing$$

$$y_o = \varnothing; \;\; y_o \in \{\mathbb{T}, \mathbb{F}\} \text{ by definition of } s^+$$

$\therefore O_S[n] = 0$ contradicts the definition of solution s^+ of a decision problem

$$O_S[n] \geqslant 0 \text{ by definition of magnitude}$$

$$\therefore O_S[n] > 0$$

4.7 $O_T[n] > 0^*$

Assuming Program is not void

4.7.1 Proof

Assume $O_T[n] = 0$

$$O_T[n] := |\mathcal{L}|$$

$$O_T[n] = 0 \Rightarrow y_o \notin \{\mathbb{T}, \mathbb{F}\}$$

$$y_o \notin \{\mathbb{T}, \mathbb{F}\};\ \ y_o \in \{\mathbb{T}, \mathbb{F}\} \text{ by definition of } s^+$$

$\therefore O_T[n] = 0$ contradicts the definition of solution s^+ of a decision problem

$$O_T[n] \geqslant 0 \text{ by definition of magnitude}$$

$$\therefore O_T[n] > 0$$

4.8 $O[n] > 0^*$

Assuming Program is not void

4.8.1 Proof

$$O[n] := O_T[n] + O_S[n]$$

$$O_T[n] > 0;\ \ O_S[n] > 0$$

$$\therefore O[n] > 0$$

4.9 $O[n] > O_T[n]_*$

Assuming Program is not void

4.9.1 Proof

$$O[n] := O_T[n] + O_S[n]$$

$$O_S[n] > 0$$

$$\therefore O[n] > O_T[n]$$

4.10 $O[n] > O_S[n]^*$

Assuming Program is not void

4.10.1 Proof

$$O[n] := O_T[n] + O_S[n]$$

$$O_T[n] > 0$$

$$\therefore O[n] > O_S[n]$$

4.11 $O[n+1] \geqslant O[n]$

4.11.1 Proof

$$X_i = \{x_1, ..., x_n, C\}; \;\; \hat{X}_i = \{x_1, ..., x_{n+1}, C\}$$

$$O[n] = |\, s^+[X_i]|$$
$$O[n+1] = \hat{O}[n] = |\, s^+[\hat{X}_i]|$$

For general solutions s^+

$$s^+[\hat{X}_i] \supseteq s^+[X_i]$$
$$\Rightarrow |s^+[\hat{X}_i]| \geqslant |s^+[X_i]|$$
$$\therefore \hat{O}[n] = O[n+1] \geqslant O[n]$$

5 Worst Case Complexity $O_{wc}[n]$

6 Polynomial Complexity

6.1 Definition

Decision problem D with solution s^+ has polynomial total complexity $O[n]$ if

$$\exists K, C, \lambda_1 ... \lambda_K \quad :$$
$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n$$

6.2 Polynomial Problems

Define $\mathbb{P}$, the set of Decision Problems that can be solved with Polynomial Complexity

$$\mathbb{P} := \{D_1, D_2, ...\} :$$

$$\exists K, C, \lambda_1 ... \lambda_K :$$
$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n, D_i \in \mathbb{P}$$

6.3 Polynomial Order of Complexity

Solution s^+ with total complexity $O[n]$ is said to be of order n^K

$$O[n] \sim n^K$$
$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n$$

6.4 Property of Polynomial Complexity 1

Solutions with polynomial complexity have convergent complexity

$$lim_{n \to \infty} \frac{O[n+1]}{O[n]} = 1$$

6.4.1 Proof

$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C$$
$$O[n+1] = (\lambda_K (n+1))^K + (\lambda_{K-1} (n+1))^{K-1} ... + \lambda_1 (n+1) + C$$
$$= (\lambda_K n)^K + (\tilde{\lambda}_{K-1} n)^{K-1} ... + \tilde{\lambda_1} n + \tilde{C}$$

$$lim_{n \to \infty} \frac{O[n+1]}{O[n]}$$

$$= lim_{n \to \infty} \frac{(\lambda_K n)^K + (\tilde{\lambda}_{K-1} n)^{K-1} ... + \tilde{\lambda_1} n + \tilde{C}}{(\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C}$$

$$= lim_{n\to\infty} \frac{(\lambda_K n)^K}{(\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C} + \frac{(\tilde{\lambda}_{K-1} n)^{K-1}}{(\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C} + ... + \frac{\tilde{\lambda}_1 n}{(\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C} + \frac{\tilde{C}}{(\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C}$$

$$= 1 = lim_{n\to\infty} \frac{O[n+1]}{O[n]}$$

6.5 Property of Polynomial Complexity 2

$$\exists K, \hat{C}, \hat{\lambda}_1, ..., \hat{\lambda}_{K-1} :$$
$$O[n+1] - O[n] = f_{n+1}[n] = (\hat{\lambda}_{K-1} n)^{K-1} ... + \hat{\lambda}_1 n + \hat{C} \quad \forall n$$

6.5.1 Proof

$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C$$
$$O[n+1] = (\lambda_K (n+1))^K + (\lambda_{K-1} (n+1))^{K-1} ... + \lambda_1 (n+1) + C$$
$$= (\lambda_K n)^K + (\tilde{\lambda}_{K-1} n)^{K-1} ... + \tilde{\lambda_1} n + \tilde{C}$$
$$O[n+1] - O[n] = ((\tilde{\lambda}_{K-1} - \lambda_{K-1}) n)^{K-1} ... + (\tilde{\lambda_1} - \lambda_1) n + (\tilde{C} - C)$$
$$O[n+1] - O[n] = (\hat{\lambda}_{K-1} n)^{K-1} ... + \hat{\lambda}_1 n + \hat{C}$$

6.6 Property of Polynomial Complexity 3

$$limit_{n\to\infty} \frac{f_{n+1}[n]}{O[n]} = 0$$

6.6.1 Proof

$$limit_{n\to\infty} \frac{O[n+1]}{O[n]} = 1$$
$$limit_{n\to\infty} \frac{O[n] + f_{n+1}[n]}{O[n]} = 1$$
$$limit_{n\to\infty} \frac{O[n]}{O[n]} + \frac{f_{n+1}[n]]}{O[n]} = 1$$
$$limit_{n\to\infty} 1 + \frac{f_{n+1}[n]]}{O[n]} = 1$$
$$limit_{n\to\infty} \frac{f_{n+1}[n]]}{O[n]} = 0$$

6.7 Total Polynomial Complexity Implies Time bounded Polynomial Complexity

$$D \in \mathbb{P} \Longrightarrow O_T[n] < (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C$$

6.7.1 Proof

$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \;\; \forall n$$
$$O[n] := O_T[n] + O_S[n]; \;\; O_S[n] > 0$$
$$\therefore O_T[n] < (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \;\; \forall n$$

6.8 Total Polynomial Complexity Implies Space bounded Polynomial Complexity

$$D \in \mathbb{P} \Longrightarrow O_S[n] < (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C$$

6.8.1 Proof

$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \;\; \forall n$$
$$O[n] := O_T[n] + O_S[n]; \;\; O_T[n] > 0$$
$$\therefore O_S[n] < (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \;\; \forall n$$

6.9 Polynomial Complexity in Space and Time Implies Polynomial Total Complexity

$$(O_S[n] == (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + \lambda_0)$$
$$\wedge$$
$$(O_T[n] == (\hat{\lambda}_M n)^M + (\hat{\lambda}_{M-1} n)^{M-1} ... + \hat{\lambda}_1 n + \hat{\lambda}_0)$$
$$\Rightarrow D \in \mathbb{P}$$

6.9.1 Proof

$$O_S[n] = \lambda_K n^K + \lambda_{K-1} n^{K-1} + ... + \lambda_1 n + \lambda_0$$
$$O_T[n] = \hat{\lambda}_M n^M + \hat{\lambda}_{M-1} n^{M-1} + ... + \hat{\lambda}_1 n + \hat{\lambda}_0$$
$$O[n] = O_S[n] + O_T[n]$$
$$^*O[n] = (\hat{\lambda}_0 + \lambda_0) + n(\lambda_1 + \hat{\lambda_1}) + ... + n^K(\lambda_K + \hat{\lambda_K}) + \hat{\lambda}_{K+1} n^{K+1} + ... + \hat{\lambda}_M n^M$$
$$\therefore O[n] \text{ has polynomial total complexity by definition}$$

* Assume K < M, similar proof for K=M, K>M

7 Non-Polynomial Complexity

7.1 Definition

Decision problem $\tilde{D}$ with solution s^+ has non-polynomial total complexity $O[n]$ if

$$\nexists K, \lambda_0, \lambda_1, ..., \lambda_K \ :$$
$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n$$

7.2 Non-Polynomial Problems

Define $\mathcal{N}$, the set of Decision Problems that cannot be solved with Polynomial Complexity

$$\mathcal{N} := \{\tilde{D}_1, \tilde{D}_2, ...\} :$$
$$\nexists K, C, \lambda_1 ... \lambda_K \ :$$
$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n, \ s^+ \in S_i^+, \ \tilde{D}_i \in \mathcal{N}$$

7.3 $\mathbb{P}$ and $\mathcal{N}$ are disjoint

$$\mathbb{P} \cap \mathcal{N} = \varnothing$$

7.3.1 Proof

Let D $\in \mathcal{N}$

$$\nexists K, C, \lambda_1 ... \lambda_K \ :$$
$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n$$

Assume D $\in \mathbb{P}$

$$\exists K, C, \lambda_1 ... \lambda_K \ :$$
$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n$$

Contradicts the definition of $\mathcal{N}$

$$\therefore D \in \mathcal{N} \Rightarrow D \notin \mathbb{P}$$

Let D $\in \mathbb{P}$

$$\exists K, C, \lambda_1 ... \lambda_K \ :$$
$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n$$

Assume D $\in \mathcal{N}$

$$\nexists K, C, \lambda_1 ... \lambda_K \ :$$
$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n$$

Contradicts the definition of $\mathbb{P}$

$$\therefore D \in \mathbb{P} \Rightarrow D \notin \mathcal{N}$$

$$D \in \mathcal{N} \Rightarrow D \notin \mathbb{P}; D \in \mathbb{P} \Rightarrow D \notin \mathcal{N}$$
$$\therefore \mathbb{P} \cap \mathcal{N} = \varnothing$$

8 Divergent Complexity

See Appendix for additional proofs on Convergent and Divergent Functions

9 Divergent Programs

See Appendix for additional proofs on Convergent and Divergent Functions

10 Problems

10.1 Alternate Terminology

A problem is also referred to as a "Question" - a bounded boolean expression

10.2 True False Question

Also referred to as a "Decision Problem"

10.3 Numerical Question

10.4 System Question

11 Solution to a Question

11.1 Worst Case Complexity of Solution s^+

11.2 Solution Space of a Question

11.3 Candidate Solutions

12 Subfunctions

13 Subfunction Decomposition of Program

14 Subsolutions

15 Inductive Complexity Expressions

$$O[n] = O_T[n] + O_S[n]$$
$$O[n+1] = O_T[n+1] + O_S[n+1]$$

$$f_{n+1}[n] := O[n+1] - O[n]$$

15.1 Inductive Space and Time Formulas

$$f^T_{n+1}[n] := O_T[n+1] - O_T[n]$$
$$O_T[n+1] = O_T[n] + f^T_{n+1}[n]$$
$$f^S_{n+1}[n] := O_S[n+1] - O_S[n]$$
$$O_S[n+1] = O_S[n] + f^S_{n+1}[n]$$

15.2 Inductive Function Expressions

Relate $f_{n+1}[n]$ to equivalence functions

$$O[n] = O_T[n] + O_S[n]$$
$$O[n+1] = O_T[n+1] + O_S[n+1] = O[n] + f_{n+1}[n]$$
$$O_T[n] = O[n] - O_S[n]$$
$$O_S[n] = O[n] - O_T[n]$$

$$f_{n+1}[n] = O[n+1] - O[n]$$
$$f_{n+1}[n] = O_T[n+1] + O_S[n+1] - O[n]$$
$$f_{n+1}[n] = O_T[n+1] - O_T[n] + O_S[n+1] - O_S[n]$$
$$f_{n+1}[n] = O[n+1] - O_T[n] - O_S[n]$$
$$f_{n+1}[n] = f^T_{n+1}[n] + f^S_{n+1}[n]$$

15.3 Zero Order Space Inductive Function

$$Let\ \ O_S[n] \sim n^0$$

$$f_{n+1}[n] = O_T[n+1] - O_T[n] + O_S[n+1] - O_S[n] = O_T[n+1] - O_T[n]$$

16 Discrete Derivative; Z Transform

16.1 Discrete Derivative

Define derivative for discrete function f[n]

$$\Delta_n^1 f[n] := f[n+1] - f[n]$$

We will use the above definition for the remainder of this document

16.2 Zero Order Derivative

$$\Delta_n^0 f[n] = f[n]$$

16.3 Kth Discrete Derivative

Define the Kth derivative of discrete function $f[n]$

$$\Delta_n^K f[n] := \Delta_n^{K-1} f[n+1] - \Delta_n^{K-1} f[n]$$

16.4 Kth Discrete Derivative as an Alternating Sum

$$\Delta_n^K f[n] := \Delta_n^{K-1} f[n+1] - \Delta_n^{K-1} f[n]$$

$$= (\Delta_n^{K-2} f[n+2] - \Delta_n^{K-2} f[n+1]) - (\Delta_n^{K-2} f[n+1] - \Delta_n^{K-2} f[n])$$

$$= (\Delta_n^{K-2} f[n+2] - 2\Delta_n^{K-2} f[n+1] - \Delta_n^{K-2} f[n])$$

$$= \textstyle\sum_{i=0}^{K} (-1)^j \ ({}_K C_j) \ \Delta_n^0 f[n+j]$$

$$= \textstyle\sum_{i=0}^{K} (-1)^j \ ({}_K C_j) \ f[n+j]$$

16.5 Z Transform

Define the Z Transform for discrete function f[n]

$$\mathcal{Z}(f[n]) := \textstyle\sum_{n=0}^{\infty} f[n] z^{-n}$$

16.6 Z Transform of 0 Order Derivative

$$\Delta_n^0 f[n] := f[n]$$

$$\mathcal{Z}(\Delta_n^0 f[n]) = \mathcal{Z}(f[n])$$

16.7 Z Transform of 1st Derivative

$$\Delta_n^1 f[n] := f[n+1] - f[n]$$

$$\mathcal{Z}(\Delta_n^1 f[n]) = \mathcal{Z}(f[n+1] - f[n])$$

$$= \sum_{n=0}^{\infty}(f[n+1] - f[n])z^{-n}$$

$$= \sum_{n=0}^{\infty}(f[n+1]z^{-n} - f[n]z^{-n})$$

$$= \sum_{n=0}^{\infty} f[n+1]z^{-n} - \sum_{n=0}^{\infty} f[n]z^{-n}$$

$$= \sum_{m=0}^{\infty} f[m+1]z^{-m} - \sum_{n=0}^{\infty} f[n]z^{-n}$$

Let

$$\hat{m} = m + 1; \;\; m = \hat{m} - 1$$

$$= \sum_{m=0}^{\infty} f[\hat{m}]z^{-(\hat{m}-1)} - \mathcal{Z}(f[n])$$

$$= z^1 \sum_{\hat{m}=1}^{\infty} f[\hat{m}]z^{-\hat{m}} - \mathcal{Z}(f[n])$$

$$= z^1 \sum_{\hat{m}=1}^{\infty} f[\hat{m}]z^{-\hat{m}} + f[0] - f[0] - \mathcal{Z}(f[n])$$

$$= z^1 \sum_{\hat{m}=0}^{\infty} f[\hat{m}]z^{-\hat{m}} - f[0] - \mathcal{Z}(f[n])$$

$$= z^1 \mathcal{Z}(f[n]) - f[0] - \mathcal{Z}(f[n])$$

$$\mathcal{Z}(\Delta_n^1 f[n]) = \mathcal{Z}(f[n])(z^1 - 1) - f[0]$$

16.8 Z Transform of Kth Derivative

$$\mathcal{Z}(f[n]) := \sum_{n=0}^{\infty} f[n]z^{-n}$$

$$\mathcal{Z}(\Delta_n^K f[n]) = \sum_{n=0}^{\infty} \Delta_n^K f[n]z^{-n}$$

$$= \sum_{n=0}^{\infty} \sum_{i=0}^{K} (-1)^j \; ({}_K C_j) \; f[n+j]z^{-n}$$

$$= \sum_{n=0}^{\infty}(f[n+K] - ({}_K C_1)f[n+K-1] + ({}_K C_2)f[n+K-2] - ... \pm f[n])z^{-n}$$

$$= \sum_{n=0}^{\infty} f[n+K]z^{-n} - ({}_K C_1)f[n+K-1]z^{-n} + ({}_K C_2)f[n+K-2]z^{-n} - \\ ... \pm f[n]z^{-n}$$

$$= z^K \mathcal{Z}(f[n]) + \sum_{i=0}^{K-1} f[i] - ({}_K C_1)z^{K-1}\mathcal{Z}(f[n]) - \sum_{j=0}^{K-2} f[j] + \\ ({}_K C_2)z^{K-2}\mathcal{Z}(f[n]) + \sum_{k=0}^{K-3} f[k] - ... \pm \mathcal{Z}(f[n])$$

When K is odd

$$\mathcal{Z}(\Delta_n^K f[n]) = (z-1)^K \mathcal{Z}(f[n]) + \sum_{i=0}^{\frac{n+1}{2}} f[2i] \quad K > 0$$

When K is even

$$\mathcal{Z}(\Delta_n^K f[n]) = (z-1)^K \mathcal{Z}(f[n]) + \sum_{j=0}^{\frac{n}{2}} f[2j+1] \quad K > 0$$

Computation Basis

17 Definition of Basis

18 Definition of Canonical Basis

19 Convertible Instructions

20 Theorem of Polynomial Subfunctions

21 Theorem of Divergent Subfunctions

22 Two integer Sum to N

23 Knapsack Problem

Parallel Programs

24 Workers

25 Definition of a Parallel Program

Optimal Programs

26 Optimal Complexity

27 Optimal Solutions

Appendix

28 Domains

29 Bounded Input Bounded Output

In abstraction theory, functions have a domain (bounded set of every possible input and output)

Convergence, Divergence, Divergent Complexity

30 Convergent Functions

31 Divergent Functions

References

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